Time-domain features: 20 equations

PCG-BP analysis version 2 | Feature calculation specification

For each complete adjacent S1-S2-S1 cycle, a, b, c, d, e denote S1 onset, S1 offset, S2 onset, S2 offset and next S1 onset, in zero-based recording samples. They must satisfy a < b < c < d < e. Sampling frequency is f_s. States use half-open sample intervals [a,b), [b,c), [c,d), [d,e).

Only complete cycles contained in a segment contribute. The next S1 onset may equal the exclusive segment end; its waveform belongs to the next segment. Detection is performed once over the whole recording.

Intervals are in seconds; ratios are dimensionless. RR is a PCG cycle-period surrogate from consecutive S1 onsets, not an ECG-measured RR. T_S1, T_Sys, T_S2 and T_Dia denote IntS1, IntSys, IntS2 and IntDia. All ten quantities use the same complete-cycle pool.

$$RR_i = T_{S1,i} + T_{Sys,i} + T_{S2,i} + T_{Dia,i}$$

Per-cycle quantity	Equation	Output columns
RR	$(e_i - a_i)/f_s$	m_RR sd_RR
IntS1	$(b_i - a_i)/f_s$	m_IntS1 sd_IntS1
IntS2	$(d_i - c_i)/f_s$	m_IntS2 sd_IntS2
IntSys	$(c_i - b_i)/f_s$	m_IntSys sd_IntSys
IntDia	$(e_i - d_i)/f_s$	m_IntDia sd_IntDia
Ratio_SysRR	$T_{Sys,i}/RR_i$	m_Ratio_SysRR sd_Ratio_SysRR
Ratio_DiaRR	$T_{Dia,i}/RR_i$	m_Ratio_DiaRR sd_Ratio_DiaRR
Ratio_SysDia	$T_{Sys,i}/T_{Dia,i}$	m_Ratio_SysDia sd_Ratio_SysDia
Ratio_S1RR	$T_{S1,i}/RR_i$	m_Ratio_S1RR sd_Ratio_S1RR
Ratio_S2RR	$T_{S2,i}/RR_i$	m_Ratio_S2RR sd_Ratio_S2RR

Aggregation within each segment

$$m_q = \frac{1}{K}\sum_{i=1}^{K} q_i$$

$$sd_q = \sqrt{\frac{1}{K-1}\sum_{i=1}^{K}(q_i - m_q)^2}$$

Ratios are calculated per cycle, then averaged. Sample SD uses K-1 (an explicit implementation convention). Mean is NaN for K=0; SD is NaN for K<2. Invalid or ambiguous cycles are excluded, never bridged or filled with zero.

Source: Tang et al. (2018), PCG Classification Using Multidomain Features and SVM Classifier, Tables 1-2 and Sections 2.3-2.5.3. DOI: 10.1155/2018/4205027. Feature definitions follow the paper; this project retains its 30-500 Hz Butterworth filter and timing-based detector rather than the paper's 10 Hz high-pass, spike removal and Springer HSMM.